Optimising Weak Reference Processing in the JVM Z Garbage Collector

Master's Thesis in Computer Science

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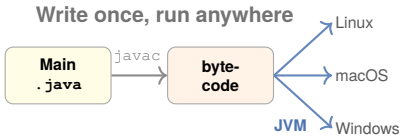
Agenda

Background

Java and the JVM

- One of the most used programming languages in the world
- Write once, compile once – runs on *any* machine via the **JVM**
- **HotSpot JVM** is the most popular implementation (Oracle / OpenJDK)

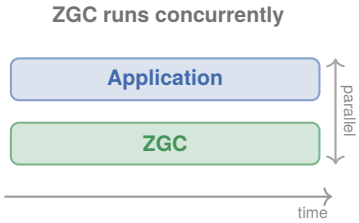


Garbage Collection

- Java has **automatic memory management** → no `free()` or `delete`
- The GC periodically finds objects the application can no longer reach and reclaims their memory

ZGC

- **Concurrent**: runs *alongside* the application → very short stop-the-world pauses
- **Relocating**: moves live objects to new compact regions. Old memory reused



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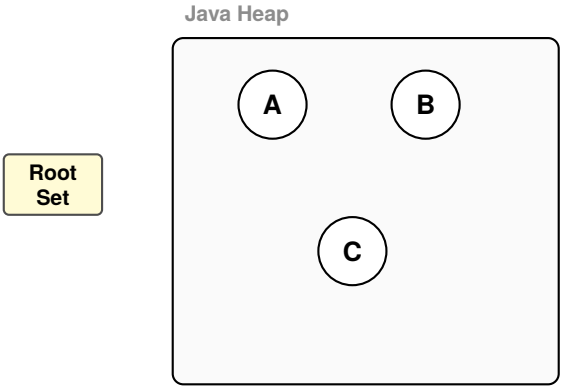
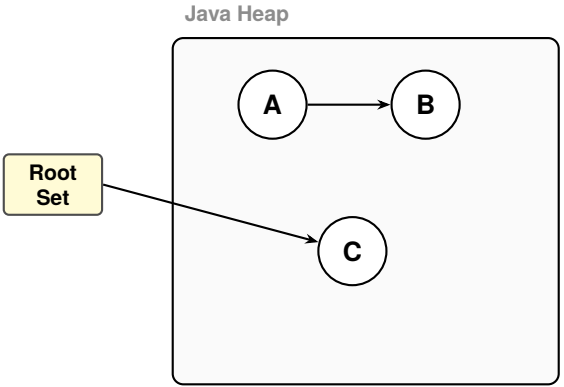


- Objects live on the **heap**, the root set is always alive

Garbage Collection



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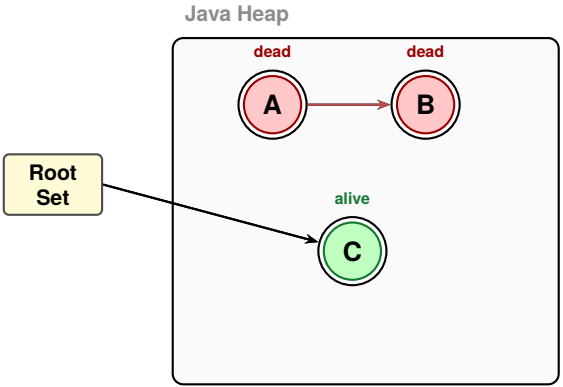
- GC **marks**: follows all references reachable from roots

Garbage Collection



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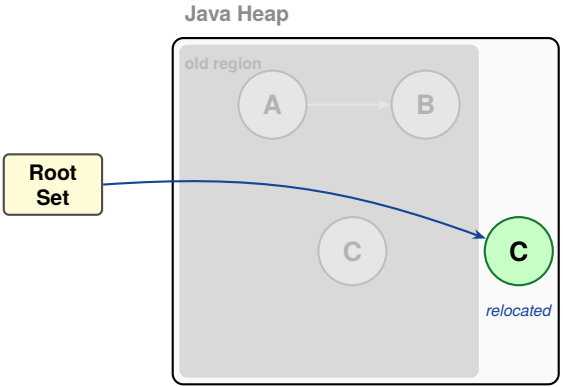
- GC **marks**: follows all references reachable from roots
- A and B are unreachable $\rightarrow$ marked as **dead**. C is reachable $\rightarrow$ marked as **alive**

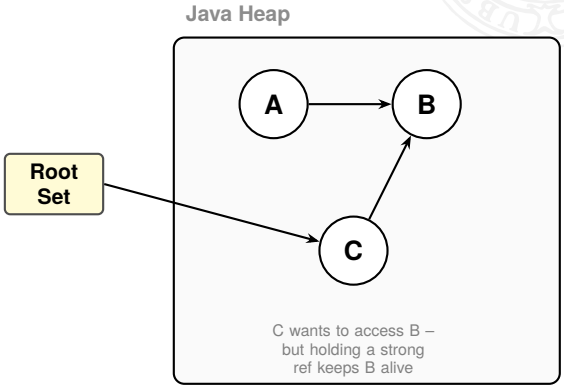
Garbage Collection



- Objects live on the **heap**, the root set is always alive
- GC **marks**: follows all references reachable from roots
- A and B are unreachable $\rightarrow$ marked as **dead**. C is reachable $\rightarrow$ marked as **alive**
- GC **relocates** C to a new region; old memory reused for future allocations

Weak References

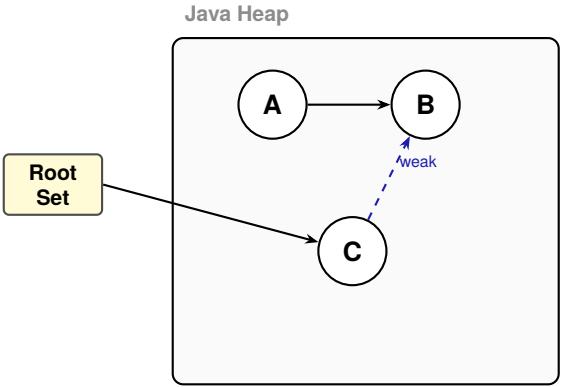
- Strong references keep objects alive. If you can reach it, it won't be collected
- But sometimes you want to **observe** an object *only while it is still alive*
- But if you hold a strong reference to the object, you keep it alive forever, defeating the purpose



The problem

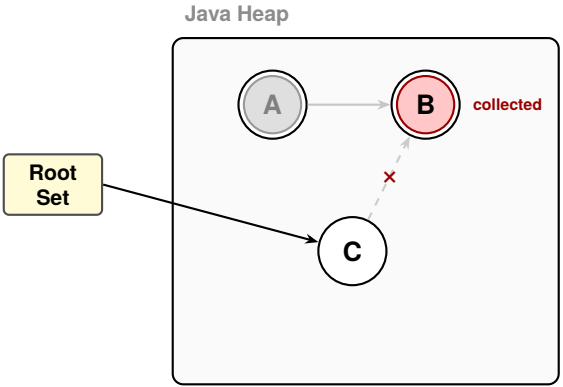
You want to hold a reference that **does not** prevent garbage collection but still lets you read the object if it hasn't been collected yet.

Weak References



- A **weak reference** does not keep an object alive since it does not count as reachability for GC purposes

Weak References



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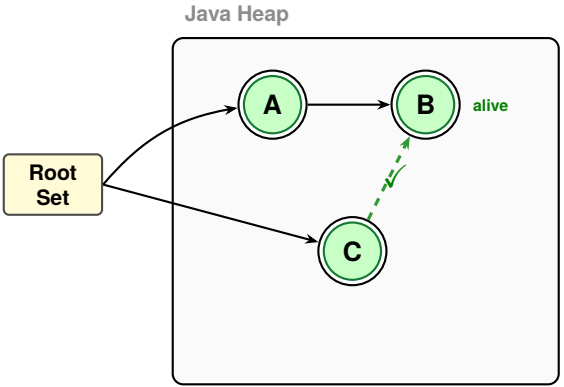
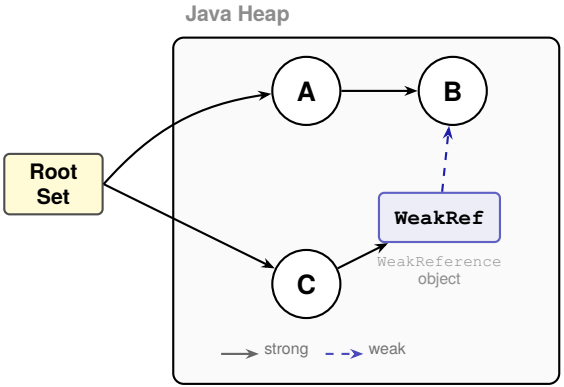
- If B has no *strong* path from roots: GC collects B; weak ref becomes `null`

Weak References



- A **weak reference** does not keep an object alive since it does not count as reachability for GC purposes
- If B has no *strong* path from roots: GC collects B; weak ref becomes `null`
- Add strong ref between root and A: B reachable through A $\rightarrow$ The weak ref from C to B is **valid**

The `WeakReference` Object

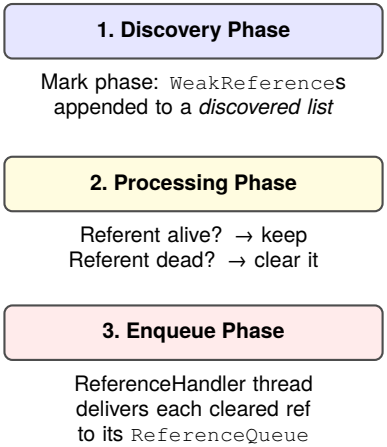


- The `WeakReference` object exists to support the `ReferenceQueue` callback
- When the GC collects B, the `WeakReference` is enqueued so the application can react

```
// With notification queue
new WeakReference<>(obj, queue)

// No queue
new WeakReference<>(obj)
```

Reference Processing Pipeline



- During marking: found `WeakReference`s go onto a linked list
- After marking: check liveness per ref, clear dead ones
- Cleared refs handed to the `ReferenceHandler` thread for queue delivery

Motivation

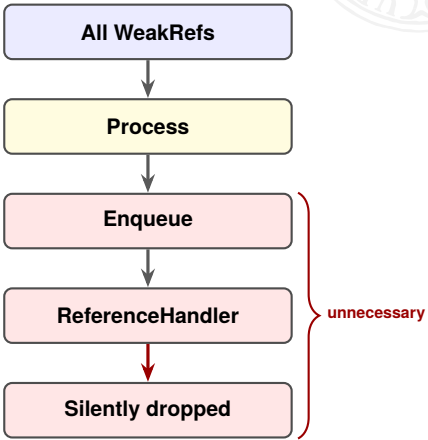
The Problem with Queue-less Weak References

- GC performance directly impacts application performance
- ZGC was designed for high performance – every unnecessary operation matters

The issue

When a `WeakReference` *without* a `ReferenceQueue` is cleared, it is still handed to the `ReferenceHandler` thread – which then silently drops it, since there is no queue.

Unnecessary work for every single queue-less weak reference.



Research Questions

RQ1 – Pipeline Optimisations

How do specific modifications to ZGC's processing of `WeakReference` objects *without* a `ReferenceQueue` affect reference-processing time, total GC cycle time, GC memory footprint, and Java heap usage?

RQ2 – Weak Fields

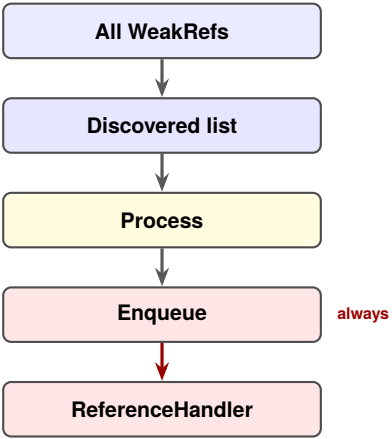
How does expressing weak semantics through annotated object fields affect reference processing time, GC cycle time, GC memory footprint, and Java heap usage compared to the optimised `WeakReference`-based approach of RQ1?

RQ1: optimise the *existing* pipeline without changing how weak refs are represented

RQ2: eliminate the `WeakReference` object entirely

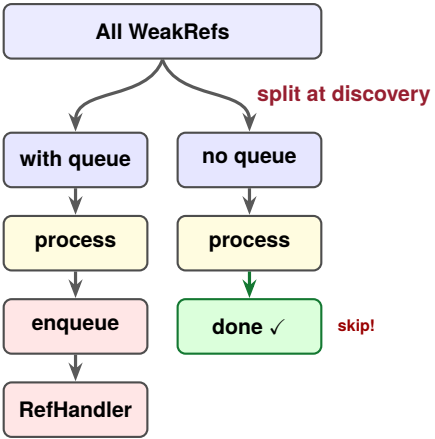
Design & Implementation

Optimisation 1: Skip Enqueue Path



- Every weak ref goes through the full pipeline
- Queue-less refs still hit the `ReferenceHandler` – which drops them
- Pure waste for weak references without a queue

Optimisation 1: Skip Enqueue Path

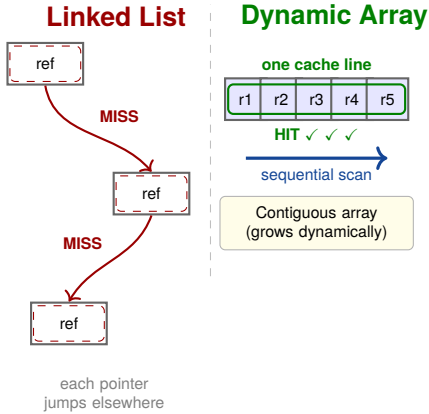


- At discovery: check the `ReferenceQueue` field
- Route queue-less refs to a *separate* discovered list
- No-queue pipeline: process (check + clear) – then **stop**
- With-queue: full existing behaviour

Why two lists instead of a branch?

A branch requires a memory load per ref to check the queue field. Two specialised loops avoid that per-reference cost.

Optimisation 2: Dynamic Array Discovered List

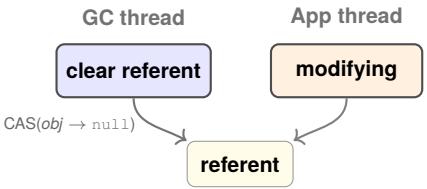


- Linked list: Nodes scattered across memory – no spatial locality
- Every node's next-pointer is likely a **cache miss**
- Dynamic Array: one cache load fills several consecutive entries
- Dynamically grow the array as needed

Trade-off

Higher auxiliary GC memory for array-based variants – explored in the Results section.

Optimisation 3: Optimised Clear Path



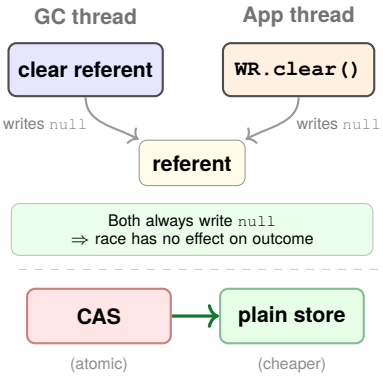
Compare-And-Swap (CAS):

```
if (*field == expected)
  *field = new_val;   // atomic
```

Guarantees only one writer wins the race.

- Clearing a referent = writing `null` to the referent field
- The GC does this with a **CAS** – an atomic instruction that only writes if the current value matches an expected value
- Needed because the ZGC runs concurrently with the application – both may access the referent field at the same time without explicit synchronisation.

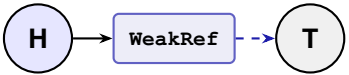
Optimisation 3: Optimised Clear Path



- **Key observation**: the only thing an app thread can write to a `WeakReference`'s referent field is `null` (via `WeakReference.clear()`)
- If a race occurs, both sides write `null` – the result is identical either way
- ⇒ The CAS is unnecessary; a **plain store** is safe

Research Question 2: Weak Fields (`@weak`)

`WeakReference` implementation



`@weak` field implementation

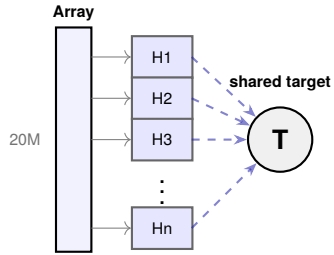
- A field annotated `@weak` is treated as a weak reference by ZGC – no `WeakReference` object needed
- Discovered during marking, stored in a dynamic array of weak fields
- Cleared the same way – but **cannot** use the optimised clear path (a regular field can be written to with any value, not just `null`)

```
class HolderClass {
    @weak TargetClass ref;
}
```

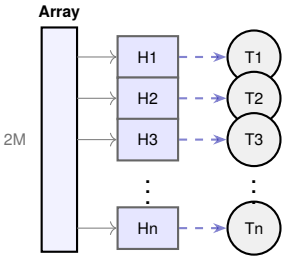
Evaluation Method

Benchmark Design

Single-Object Benchmark



Multi-Object Benchmark



- **Single**: 20M holders, one shared payload – all 20M refs processed in one GC cycle
- **Multi**: 2M holders, individual payloads, 5 gradual rounds – more realistic
- Both: no `ReferenceQueue`; both have a `@weak` field counterpart

Variants and Execution

Variant	clr	sep	dyn
None	–	–	–
Dyn	–	–	✓
Sep	–	✓	–
Sep + Dyn	–	✓	✓
Clear Path	✓	–	–
Clear Path + Dyn	✓	–	✓
Clear Path + Sep	✓	✓	–
All	✓	✓	✓
Weak Fields	–	✓	✓

- All 2^3 = 8 pipeline combinations plus Weak Fields

Execution Environment

UPPMAX supercomputer (Pelle)

AMD EPYC 9454P (Zen 4)
48 physical cores, 768 GiB RAM

4 parallel instances per run
Each: 10 JVM cores + 2 aux cores

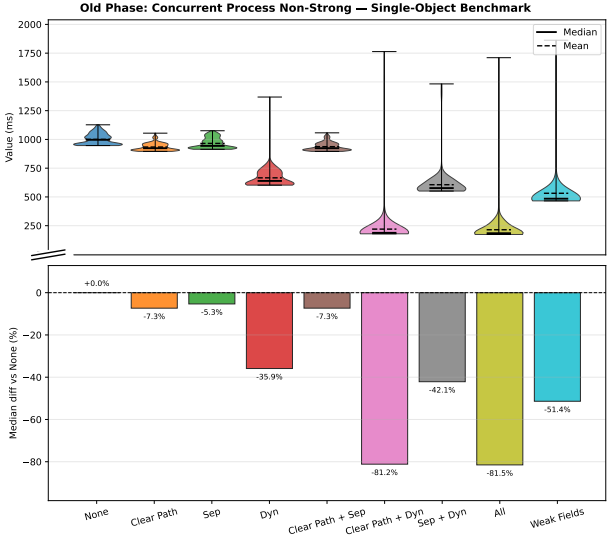
100 GB heap `-XX:+UseZGC`
250 iterations (+1 warmup discarded)

Reporting

Violin + **median** histogram over the 250 iterations.

Results

Non-Strong Processing Time – Single-Object Benchmark



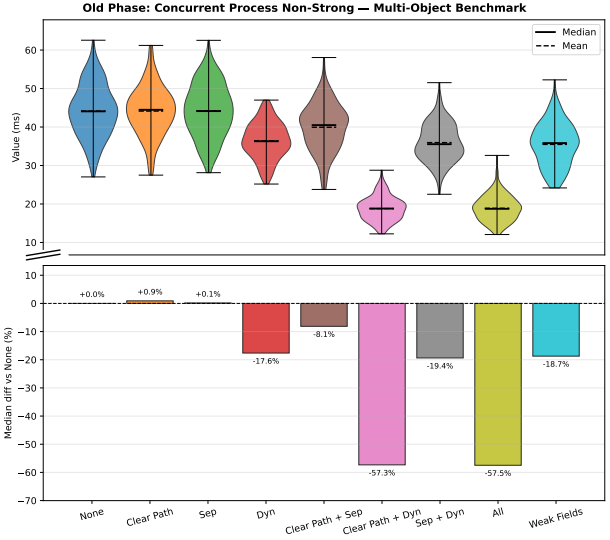
Variant	Reduction
Sep	−5%
Clear Path	−7%
Dyn	−36%
Clear Path + Dyn	**−81%**
All	**−81%**
Weak Fields	−51%

Super-additive interaction

clear_path alone: -7%
dyn alone: -36%
Together: -81%

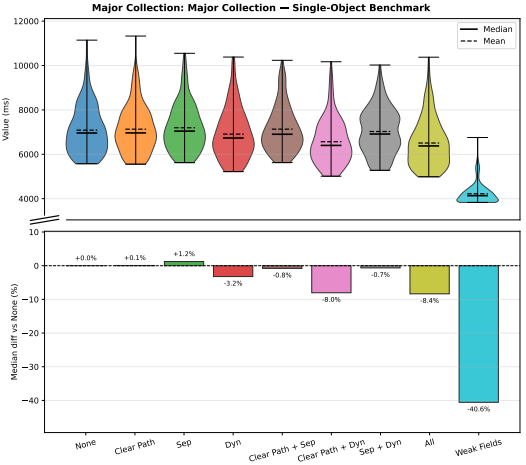
Each removes a different bottleneck.

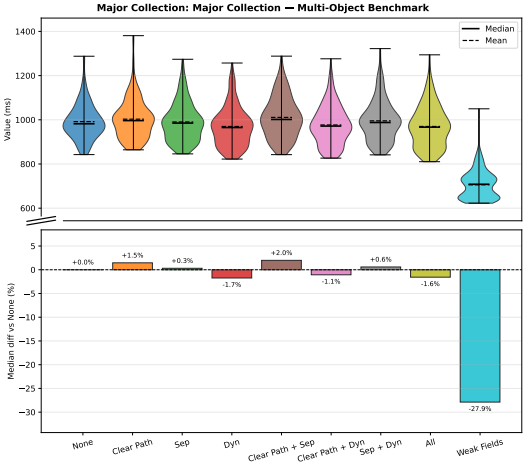
Non-Strong Processing – Multi-Object Benchmark



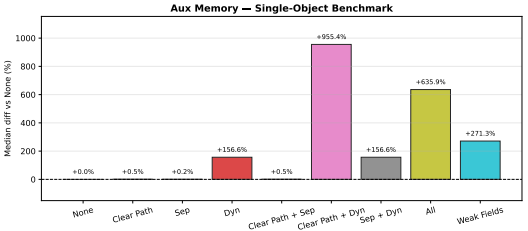
- Same pattern: Clear Path + Dyn and All at -57%
- Weak Fields: -19% – again comparable to Sep + Dyn

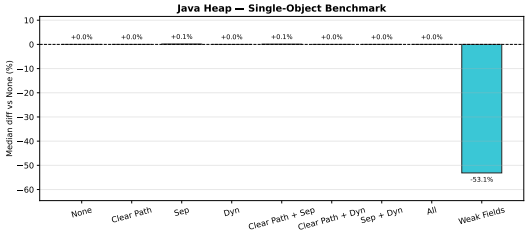
Major Collection Time





Memory Usage





- Clear Path + Dyn: $\times 10$ aux memory (1268 MB vs 120 MB)
- All: $\times 7$ (sep reduces bytes per entry vs dyn alone)

- All `WeakReference` variants: heap size identical
- Weak Fields: **-53%** heap (single), -5% (multi), no `WeakReference` objects to allocate

Conclusions

Conclusions

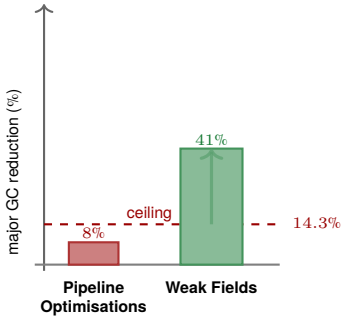
RQ1 – Pipeline Optimisations

- **81%** reduction in non-strong processing time (Clear Path + Dyn / All)
- Translates to only **8%** in total major collection time
- Hard ceiling: non-strong is 14.3% of major GC time (single) and 4.5% (multi)

RQ2 – Weak Fields

- **41%** major GC time reduction (single), **28%** (multi)
- **53%** heap reduction (single) – no `WeakReference` objects
- Structural advantage: fewer objects means less work across *every* GC phase

Key Takeaway



Optimise the pipeline (All)

-81% non-strong, but only -8% total GC
Hard ceiling: non-strong is just 14% of total GC

Change the representation (Weak Fields)

-41% total GC (single)
Fewer objects → less work in *every* GC phase

The bottleneck is the *representation*

Rethink how weak semantics is represented –
not just how to process `WeakReference`s faster.

Thank You

Questions?

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References I

Back to start

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